

Gen AI Learning Pathway: LangChain, LangGraph, LangSmith

Step 1: Foundations

- Python, APIs, JSON, basic ML concepts
- Understand LLMs (GPT, embeddings, tokens)

Step 2: Core Gen AI Concepts

- Prompt engineering
- Embeddings & Vector Databases
- RAG (Retrieval Augmented Generation)

Step 3: LangChain

- Document loaders (PDF, text)
- Text splitting
- Embeddings & vector stores (Chroma, FAISS)
- Chains & RetrievalQA

Step 4: Build RAG Apps

- Chat with PDF
- Q&A; systems
- Knowledge assistants

Step 5: LangGraph

- Stateful workflows
- Multi-step reasoning
- Build AI agents with decision flows

Step 6: LangSmith

- Debugging chains
- Observability & tracing
- Evaluation of LLM outputs

Step 7: Build Real AI Agents

- Tool usage (APIs, DBs)
- Multi-agent systems
- Autonomous workflows

Final Goal

- Production-ready AI apps
- Scalable RAG systems
- Intelligent agents with memory and tools